

CS224N: Natural Language Processing with Deep Learning

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Lecture 1 Notes - Course Overview & Word Vector Representations

Unit 1: Introduction to Word Vectors & Representation Learning

In traditional natural language processing, words were represented as discrete atomic symbols using one-hot vectors.

However, one-hot vectors suffer from the fundamental limitation of orthogonality: the dot product between any two one-hot vectors is either 0 or 1.

To overcome this limitation, distributional semantics uses dense continuous word vectors (word embeddings) in which the dot product between two word vectors can be any value between -1 and 1.

Unit 2: Continuous Bag-of-Words (CBOW) Model Architecture

Word2Vec is a computationally efficient framework introduced by Mikolov et al. for learning word embeddings from

In the Continuous Bag-of-Words (CBOW) architecture, the model predicts a target center word given a context w

Unit 3: Continuous Skip-Gram Architecture & Negative Sampling

The Continuous Skip-gram architecture operates inversely to CBOW: given a single center word, the model predicts

Standard Skip-gram uses a full Softmax over the entire vocabulary $|V|$, which becomes computationally prohibitive

Unit 4: Pointwise Mutual Information (PMI) & Matrix Factorization

Before Word2Vec, co-occurrence matrix factorization methods were widely used to learn word representations. A

Pointwise Mutual Information (PMI) quantifies the association between a word w and a context word c by measu

Unit 5: Evaluation & Intrinsic vs Extrinsic Word Vector Benchmarks

Word embedding models are evaluated using two complementary methodologies: intrinsic evaluation and extrinsic

Intrinsic evaluation measures word vector quality directly on specific subtasks without integrating vectors into an